Ashton Hendrickson

AEROSPACE ENGINEERING STUDENT

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TECHNICAL PROFILE

Aerospace Engineering student (3.71 GPA) with hands-on experience in SolidWorks CAD, Python automation, and MATLAB. Recent exposure to FAA airworthiness documentation and manufacturing processes through a job shadow at Able Aerospace (Textron Aviation). Looking for an internship in mechanical or aerospace design and manufacturing engineering.

EDUCATION

Chandler-Gilbert Community College — Chandler, AZ

Associate in Science / Engineering Transfer (Aerospace Engineering Track)

- Expected Bachelor's Graduation: Spring 2028
- GPA: 3.71 / 4.0
- Honors: Member, Phi Theta Kappa National Honor Society
- Relevant Coursework: Calculus I & II (Straight A's), Physics I — Kinematics (A), Computer-Aided Design (SolidWorks), MATLAB for Engineers

TECHNICAL SKILLS

- **CAD & 3D Modeling:** SolidWorks (Part Design, Assemblies, Rotational Mates, 2D Engineering Drawings)
- **Programming & Automation:** Python (API integration, workflow automation, computer vision), MATLAB
- **Technical Concepts:** Kinematics, Vector Calculus, Thermal Systems, Engineering Physics, Dimensional Analysis

JOB SHADOW EXPERIENCE

Able Aerospace (Textron Aviation), Engineering Specialist — *Mesa, AZ | Aug 2026*

- Shadowed an engineering specialist reviewing design documents, confirming airworthiness data, and signing off on custom repair and part documentation submitted to the FAA.
- Observed production of flight-critical hardware — rotor hubs, transmission shafts, landing gear, and wing components — on multi-axis CNC mills, shot-peening equipment, and electroplating lines.
- Saw how engineers design custom tooling and test rigs (e.g., water testing tanks) in SolidWorks in-house, and how the company tracks engineering data and workflows with an AI-assisted software tool.

ENGINEERING PROJECTS

Scratch-Built Sounding Rocket — *Independent Project — In Progress*

- Independently designing and fabricating a custom sounding rocket in SolidWorks, including a motor mount, recovery system with piston-based bay separation, and payload bay for onboard flight telemetry.
- Used OpenRocket flight simulation to redesign fin geometry after finding an under-stable margin, improving stability from 0.75 to 1.49 calibers; selected materials by failure mode (PETG over PLA for motor heat, epoxy over wood glue for bonding printed parts to the airframe).

10-Part 3D CAD Desk Fan Assembly — *Design Course Project*

- Modeled a 10-part desk fan assembly in SolidWorks, using rotational mates to simulate blade motion within the design.
- Applied parametric design and housing constraints to keep all 10 custom parts properly toleranced and interference-free.
- Generated detailed 2D engineering drawings with standard dimensioning and tolerances for manufacturing evaluation.

WORK EXPERIENCE

Technical & Automation Assistant, Mobile Hearing Solutions — *Gilbert, AZ | Jan 2026 – Present*

- Designed and built a Python/Streamlit application integrating Google Calendar and Google Maps APIs to recommend patient appointment times that minimize technician drive time across the Phoenix metro area.
- Replaced a fully manual scheduling process, iterating through four architectures over two months to reach a production system used daily.
- Built an offline, GPU-accelerated OCR pipeline using a vision-language model to extract patient and insurance data from faxed referrals.
- Solved GPU memory constraints via 4-bit quantization to run entirely locally for HIPAA-sensitive data; deployed and in active production use.

Truck Unloader & Stocker — Walmart — *Gilbert, AZ | Nov 2024 – Jan 2026*

- Unload incoming freight and stock shelves against nightly quotas in a fast-paced warehouse environment.
- Coordinate with a team to process high-volume shipments accurately under time pressure.

CERTIFICATIONS & HONORS

MATLAB Onramp Certificate — MathWorks | Member — Phi Theta Kappa National Honor Society